Run report: genus-5 screen for étale covers at K-409 and K-571 (lead L8)

run-20261103T2209Z · study `research/ecdlp_leads_20261018` · commit `579d02b3a`

Kind. Exact enumeration of two families, an exact recount of a published list for the third, and a screen on a necessary condition, run with the native Rust crate `etale_search` (`genus5.rs`). It is not a timing or statistical measurement. Host: shared cloud container (Xeon 2.80 GHz, 4 cores, rustc 1.97.0), isolation L0. It builds on run-20261018T2144Z, which closed K-283 at genus 4.

Why genus 5. The L(1/2) budget (derived) allows $g_0 \le 5$ for K-409 and ≤ 7 for K-571, and genus ≤ 4 is closed for both. At genus 5 the Weil bound $\left(1+\sqrt{2}\right)^{10} \approx 6726$ on $\#J_B(\mathbb{F}_2)$ exceeds 409 and 571, so the bound no longer closes it and the screen has to. An $\mathbb{F}_2$-rational, geometrically $\mathbb{Z}/n$-Galois cover with Frobenius acting by λ exists iff $n \mid P_B(\lambda)$, and T_n in the Prym needs ord $\lambda \le g_0 - 1 = 4$. The run tests λ of order 1, 2, 3 and 4 (for 409 all four occur; for 571 orders 1, 2, 3).

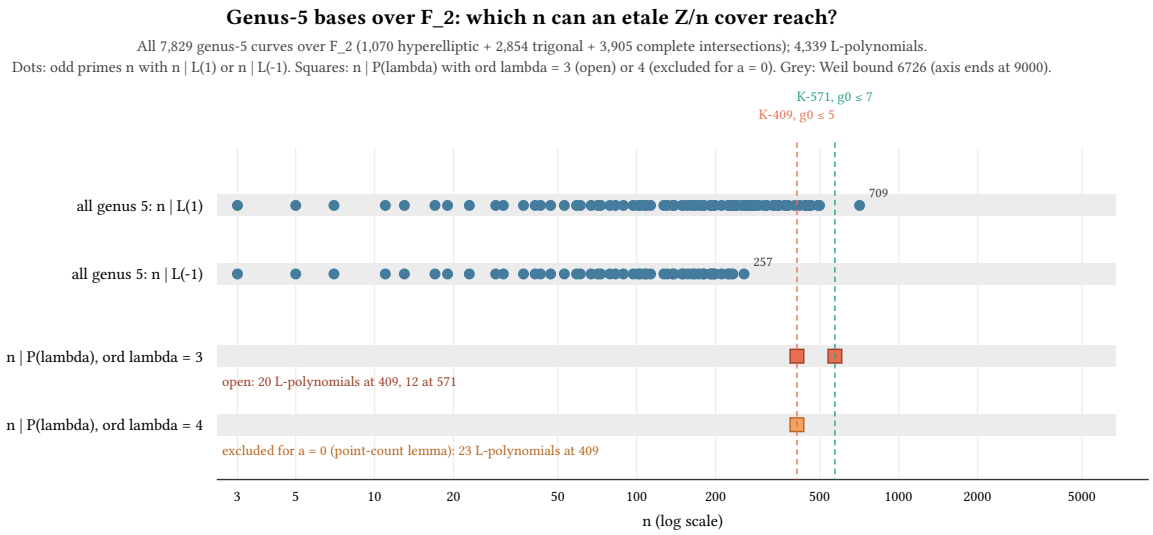
What ran. Every genus-5 curve over $\mathbb{F}_2$ is hyperelliptic, trigonal (a plane quintic with one singular point of delta invariant 1) or a complete intersection of three quadrics in $\mathbb{P}^4$ (sourced: Dragutinović, arXiv:2202.07809). That paper lists all three families with point counts and automorphism sizes. The lists were read as untrusted data; none of their code was run.

- **Hyperelliptic, independent:** every model $y^2 + qy = p$, $\deg q \le 6$, $\deg p \le 12$ (2^{20} models).
- **Trigonal, independent:** every quintic singular at $(0:0:1)$ (2^{18} forms), one node or cusp, no other singular point over $\mathbb{F}_{2^k}$, $k \le 6$.
- **Sourced recount:** every published model recounted over $\mathbb{F}_2, \ldots, \mathbb{F}_{64}$ (hyperelliptic and trigonal also $\mathbb{F}_{128}$); L from $N_1..N_5$; compared with the published counts; Weil bounds; N_6 (N_7) out of sample.
- **Not re-tested:** smoothness of the complete intersections (carried by the published Sage check).

quantity	hyperelliptic	trigonal	complete int.	all genus 5
curves (published)	1,070	2,854	3,905	**7,829**
counts = published	1,070	2,854	3,905	all
Weil / N_6 failures	0	0	0	0
distinct L	841	2,036	2,827	**4,339**
$\sum 1/\lvert\mathrm{Aut}\rvert$ (published)	512	2,817	3,584	**6,913**
independent enumeration	393,216 models	67,608 forms at one point	–	
divided by group order	$393216/768 = 512$	$7 \cdot 67608/168 = 2817$	–	
L-set = published	841 = 841	2,036 = 2,036	–	
largest $L(1)$	342	710	975	975
largest $\lvert L(-1)\rvert$	342	176	392	392
$n \mid L(\pm 1)$, 409, 571	none	none	none	**none**

Census check (sourced and measured). The union has 4,339 distinct L-polynomials, the count the paper gives. The stack counts $512 = 2^9$, $2817 = 2^{11} + 2^{10} - 2^8 + 1$ and $3584 = 2^{12} - 2^9$ add to $\lvert M_5(\mathbb{F}_2)\rvert = 6913$. The hyperelliptic and trigonal families are measured independently here. The complete-intersection family is sourced: its completeness is the paper's Theorem 4.1 plus the mass 3,584; this run checks its counts, Weil bounds and N_6, not its smoothness or isomorphism classes.

n	ord λ	L-polys	curves	$\#J(\mathbb{F}_2)$	status
409	1, 2	0	0	–	**closed**
409	3	20	40	10 – 166	**open**
409	4	23	44	6 – 249	excluded (lemma)
571	1, 2	0	0	–	**closed**
571	3	12	19	8 – 122	**open**



Orange dashed: g0 = 5 is the last base genus the budget allows (order-3 candidates remain). Teal dashed: g0 = 6, 7 also allowed, untested. Run run-20261103T2209Z.

Figure 1: Odd primes dividing $L_B(\pm 1)$ over every genus-5 curve over $\mathbb{F}_2$, and the n with $n \mid P_B(\lambda)$ for λ of order 3 and 4, against the deployed sizes whose budget allows $g_0 \geq 5$. Recorded values from `reach.csv`.

Order 4 (derived). The point-count lemma of the genus-4 run applies at $k = g_0 - 1 = 4$: if T_n is in the Prym, $\#D(\mathbb{F}_{16}) = \#B(\mathbb{F}_{16}) + 4t_4 \equiv 0 \bmod 409$, and $409 > 2^4 + 1 + 10 \cdot 2^2 + 4|t_4| = 61$ forces $\#B(\mathbb{F}_{16}) = -4t_4$. For $a = 0$, $t_4 = 1 > 0$, so $\#B(\mathbb{F}_{16}) < 0$: impossible.

Order 3 is not excluded. At $g_0 = 5$ the Prym's eigenvalues are only known to contain the μ_3-closure of T_n's, which fills $3(n-1)$ of $4(n-1)$ dimensions. The remaining $n-1$ dimensions are free, so $\#D(\mathbb{F}_8) = \#B(\mathbb{F}_8) + 3t_3 - 3s'$ with s' an unknown integer, and no point count rules the candidates out. $n \mid \#J(\mathbb{F}_8)$ holds for all 32 order-3 classes, as it must.

What this closes. No genus-5 curve over $\mathbb{F}_2$ has an étale $\mathbb{Z}/409$ or $\mathbb{Z}/571$ cover with $\lambda = \pm 1$, and the order-4 twist is excluded at $n = 409$. K-409 stays open at $g_0 = 5$ through 20 order-3 classes (40 curves), and K-571 through 12 classes (19 curves); K-571 is also open at $g_0 = 6, 7$. No security claim changes and no index-calculus cost moves.

Next. For the 59 candidate curves compute the twisted L-function over $\mathbb{F}_8$ (degree 8) for a character of $J_B(\mathbb{F}_8)[n]$ and test for $(1 - c\pi^3 X)(1 - c\bar{\pi}^3 X)$.

Records. `RUN.txt` (commits, host, SHA-256 of outputs and inputs), `g5_hyp_enum.jsonl`, `g5_trig_enum.jsonl`, `g5_sourced.jsonl`, `candidates.csv`, `reach.csv` (via `tables.awk`), `reach.plot.awk`.